

Honours Analysis 1 Tutorials (Sec. 003)

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Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

ANNOUNCEMENT: I have moved my Thursday office hours to the 4:00pm-5:00pm slot. Please let me know if this causes any scheduling issues.

1 Ninth Tutorial

Since this is the last tutorial before the second midterm exam, we will spend sometime reviewing topics in point-set topology and sequences. As for the more recent material covered (e.g. Cauchy sequences), we will focus on solving some of the ungraded homework questions.

1.1 Accumulation Points

Example 1. Consider the sequence (x_n) defined by

$$x_n := \frac{(-1)^n n}{2n - 1}.$$

We will calculate all the accumulation points of (x_n) . First, we note that the subsequences (x_{2n}) and (x_{2n+1}) are convergent with limits

$$\lim x_{2n} = \lim \left(\frac{(-1)^{2n} 2n}{2(2n) - 1} \right) = \lim \left(\frac{2n}{2(2n) - 1} \right) = \frac{1}{2}$$

and

$$\lim x_{2n+1} = \lim \left(\frac{(-1)^{2n+1} (2n+1)}{2(2n+1) - 1} \right) = - \lim \left(\frac{2n}{2(2n+1) - 1} \right) = -\frac{1}{2}.$$

As an exercise, you should rigorously prove both of these statements. Now, we know that

$$\text{Acc}(x_n) \supseteq \left\{ \pm \frac{1}{2} \right\}.$$

We claim that these are the only accumulation points of (x_n) . To see this, take a subsequence (x_{n_k}) of (x_n) that converges. Note that (x_{n_k}) must contain a subsequence of either (x_{2n}) or (x_{2n+1}) . Therefore, if (x_{n_k}) converges, it converges to either $\lim x_{2n}$ or $\lim x_{2n+1}$. Hence, the only accumulation points of (x_n) are $\pm \frac{1}{2}$. It readily follows that

$$\liminf x_n = -\frac{1}{2} \quad \text{and} \quad \limsup x_n = \frac{1}{2}.$$

At the end, we will return to this example using an equivalent formulation of the limit superior, i.e.

$$\limsup(x_n) = \lim_{m \rightarrow \infty} \sup_{n \geq m} \{x_n\}.$$

Proposition 1. *Let (x_n) be a bounded sequence in $\mathbb{R}$. Then, $\lim x_n$ exists if and only if the sequence (x_n) has exactly one accumulation point.*

Proof. One direction is easy: if (x_n) converges to $x \in \mathbb{R}$, then so must any subsequence of (x_n) . In particular, x can be the only accumulation point of (x_n) .

Conversely, suppose that the sequence (x_n) has only one accumulation point, say, x . Then, every convergent subsequence (x_{n_k}) of (x_n) must converge to this point x . By way of contradiction, suppose that (x_n) does *not* converge to this point x . Then, there is some $\varepsilon_0 > 0$ such that, for every $N \in \mathbb{N}$, there exists $n \geq N$ with the property that $|x_n - x| \geq \varepsilon_0$.¹ In particular, we can find a subsequence (x_{n_k}) of (x_n) such that $|x_{n_k} - x| \geq \varepsilon_0$, for all $k \in \mathbb{N}$. Note that this subsequence is also a bounded sequence. By the Bolzano-Weierstrass theorem, we can find a subsequence $(x_{n_{k_j}})$ of our subsequence (x_{n_k}) that converges. Since $(x_{n_{k_j}})$ is then a convergent subsequence of (x_n) , it must converge to x . But this is impossible for, as a subsequence of (x_{n_k}) ,

$$|x_{n_{k_j}} - x| \geq \varepsilon_0, \quad \text{for all } j \in \mathbb{N}.$$

This proves our claim. □

¹To see this, just negate the definition of convergence to x .

Remark 1.1. The boundedness assumption in the previous proposition is necessary. Indeed, consider the sequence (x_n) in $\mathbb{R}$ given by

$$x_n = \begin{cases} 1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Clearly, $\lim x_{2n} = \lim 1 = 1$ whence 1 is an accumulation point of (x_n) . Let now (x_{n_k}) be a convergent subsequence of (x_n) . In particular, (x_{n_k}) is bounded. Now, (x_{n_k}) either contains a subsequence of (x_{2n}) , or a subsequence of (x_{2n+1}) . However, since any subsequence of (x_{2n+1}) is unbounded, (x_{n_k}) must contain a subsequence of (x_{2n}) . In particular, (x_{n_k}) contains a subsequence converging to 1. Because (x_{n_k}) is assumed to be convergent, all subsequences converge and have the same limit as (x_{n_k}) . This tells us that $\lim x_{n_k} = 1$. In short, (x_n) has a single accumulations point, i.e. 1. However, (x_n) is unbounded and clearly cannot converge.

Proposition 2. *Let (x_n) be a bounded sequence in $\mathbb{R}$. Then, (x_n) converges if and only if*

$$\liminf x_n = \limsup x_n.$$

In either case,

$$\lim x_n = \liminf x_n = \limsup x_n.$$

Proof. Let (x_n) be a convergent sequence, with limit $x \in \mathbb{R}$. By the previous proposition, (x_n) has but a single accumulation point, denoted a . Hence,

$$\liminf x_n = \limsup x_n = a.$$

But, (x_n) is a convergent subsequence of (x_n) . Therefore, $\lim x_n = x$ must be an accumulation point of (x_n) . Since a is the only such point, we conclude that $x = a$. Especially,

$$\lim x_n = \liminf x_n = \limsup x_n.$$

Conversely, suppose that $\liminf x_n = \limsup x_n$. Denote by $\text{Acc}(x_n)$ the set of all accumulation points of (x_n) . Since (x_n) is bounded, it has a convergent subsequence (why?). Thus, $\text{Acc}(x_n)$ is non-empty. If it contains two distinct elements $a_1 < a_2$, then we would have

$$\liminf x_n \leq a_1 < a_2 \leq \limsup x_n.$$

which would be a contradiction. Hence, $\text{Acc}(x_n)$ consists of a single point a . In particular,

$$a = \liminf x_n = \limsup x_n.$$

Once again invoking the previous proposition, we see that $\lim x_n$ exists. As before, (x_n) is a convergent subsequence of (x_n) and therefore $\lim x_n$ is an accumulation point of (x_n) . It follows that

$$\lim x_n = a = \liminf x_n = \limsup x_n.$$

□

1.2 Cauchy Sequences

Example 2. Let (x_n) be a sequence of real numbers and let $r \in (0, 1)$. In addition, suppose that $|x_{n+1} - x_n| < r^n$, for all $n \in \mathbb{N}$. We show that (x_n) is convergent. Based on the given information, it is not clear what $\lim x_n$ should be. Thus, our only real option is to check that (x_n) is Cauchy.²

To this end, let $\varepsilon > 0$ be given, and choose $N \in \mathbb{N}$ so large that

$$\frac{r^N}{1-r} < \varepsilon.$$

This can be done because $r \in (0, 1)$ implies $\lim r^n = 0$. Now, given any natural numbers $m > n \geq N$,

$$\begin{aligned} |x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - \cdots - x_{n+1} + x_{n+1} - x_n| \\ &\leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \\ &\leq \sum_{k=n}^{m-1} r^k \\ &= \sum_{k=0}^{m-1} r^k - \sum_{k=0}^{n-1} r^k \\ &= \frac{1-r^m}{1-r} - \frac{1-r^n}{1-r} \end{aligned}$$

²Since $\mathbb{R}$ is complete, a sequence in $\mathbb{R}$ converges if and only if it is Cauchy. The importance of this fact should not be underestimated.

$$\begin{aligned}
&= \frac{r^n - r^m}{1 - r} \\
&\leq \frac{r^n}{1 - r} \\
&\leq \frac{r^N}{1 - r}.
\end{aligned}$$

Hence, $|x_m - x_n| < \varepsilon$ whenever $m \geq n \geq N$. This shows that (x_n) is Cauchy.

Example 3. Let (x_n) be an unbounded sequence. We claim that there exists a subsequence (x_{n_k}) such that

$$\lim_{k \rightarrow \infty} \frac{1}{x_{n_k}} = 0.$$

Since (x_n) is unbounded, we know that for every $M > 0$, one can find some $n \in \mathbb{N}$ such that $|x_n| > M$. In fact, we can find *infinitely many* n such that $|x_n| > M$. Otherwise, we would have

$$|x_n| \leq M$$

for all but finitely many $n \in \mathbb{N}$. If this were the case, then the sequence (x_n) would be bounded. Now, given $k \geq 1$, we can find $n_k \in \mathbb{N}$ such that $|x_{n_k}| > k$. Having chosen x_{n_k} , we can guarantee the existence of $n_{k+1} \in \mathbb{N}$ such that

$$|x_{n_{k+1}}| > k + 1 \quad \text{and} \quad n_{k+1} > n_k.$$

In this way, we construct a subsequence (x_{n_k}) such that $|x_{n_k}| > k$, for every $k \in \mathbb{N}$. It follows that,

$$\left| \frac{1}{x_{n_k}} - 0 \right| = \frac{1}{|x_{n_k}|} < \frac{1}{k}, \quad \forall k \geq 1.$$

Hence, $\lim_{k \rightarrow \infty} \frac{1}{x_{n_k}} = 0$.

Problem 1. Let (x_n) be a sequence with $x_n \geq 0$, for all $n \in \mathbb{N}$. Suppose that $\lim_{n \rightarrow \infty} (-1)^n x_n$ exists. Prove that (x_n) converges.

Proof. Let x denote the limit of the sequence $(-1)^n x_n$. Then, all subsequences of $(-1)^n (x_n)$ converge to this point x . In particular,

$$x = \lim_{n \rightarrow \infty} (-1)^{2n} x_{2n} = \lim_{n \rightarrow \infty} x_{2n} \geq 0$$

and, since $x_n \geq 0$ for all $n \in \mathbb{N}$,

$$x = \lim_{n \rightarrow \infty} (-1)^{2n+1} x_{2n+1} = \lim_{n \rightarrow \infty} (-x_{2n+1}) \leq 0.$$

Therefore, the only possibility is to have $x = 0$. That is, $\lim (-1)^n x_n = 0$. Let now $\varepsilon > 0$ be given. Using this last property, we can find $N \in \mathbb{N}$ such that

$$|x_n| = |(-1)^n x_n - 0| < \varepsilon$$

for all $n \geq N$. This means that $\lim x_n = 0$. $\square$

1.3 Divergence of the Harmonic Series

We now consider the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n},$$

which is often called the *harmonic series*. As mentioned in previous tutorials, the infinite series above can be formally defined by the following

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{1}{n} =: \lim_{N \rightarrow \infty} S_N.$$

whenever the limit exists. Here, we are setting $S_N := \sum_{n=1}^N \frac{1}{n}$ for each $N \geq 1$. Then, the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ exists if and only if the sequence (S_N) converges. Thus, to show that the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, it is enough to show that the sequence (S_N) is not Cauchy.

Let now $N, M \in \mathbb{N}$ be given. If $M > N$, an easy calculation tells us that

$$\begin{aligned} |S_N - S_M| = S_M - S_N &= \sum_{n=1}^M \frac{1}{n} - \sum_{n=1}^N \frac{1}{n} = \sum_{n=N+1}^M \frac{1}{n} \\ &\geq \sum_{n=N+1}^M \frac{1}{M} \\ &= \frac{M - N}{M} \\ &= 1 - \frac{N}{M}. \end{aligned}$$

In particular, taking $M = 2N$ gives

$$|S_{2N} - S_N| \geq 1 - \frac{1}{2} = \frac{1}{2}, \quad \forall N \in \mathbb{N}.$$

This shows that (S_N) cannot be Cauchy, and thus does not converge.

1.4 Point-Set Topology & Sequences

Are there any questions related to point-set topology or sequences? If not, I will prove two results which offer a nice simultaneous review of both topology and sequences.

Theorem 1.1. *Let (x_n) be a sequence and fix $x \in \mathbb{R}$. The following statements are equivalent:*

1. $\lim x_n = x$;
2. *for every open set U containing x , there exists $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$.*

Proof. Suppose that $\lim x_n = x$ and let U be an open set containing x , we must show that $x_n \in U$ for all sufficiently large n . Since U is open, there exists $\varepsilon > 0$ such that $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon) \subseteq U$. On the other hand, since $x_n \rightarrow x$, we can find $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

This is precisely the statement that $x_n \in V_\varepsilon(x) \subseteq U$, for all $n \geq N$. Thus, we have proven the first implication.

Conversely, assuming 2., we must show that $\lim x_n = x$. Given $\varepsilon > 0$, we note that $V_\varepsilon(x)$ is an open set containing x . Therefore, we can find $N \in \mathbb{N}$ such that $x_n \in V_\varepsilon(x)$ for all $n \geq N$. But,

$$x_n \in V_\varepsilon(x) \iff |x_n - x| < \varepsilon.$$

Thus, for all $n \geq N$, we must have $|x_n - x| < \varepsilon$. By definition, this means that $x_n \rightarrow x$. $\square$

The above gives us an alternative (but equivalent) definition for the convergence of sequences. It is this second definition that we work with, when dealing with more general spaces.

Theorem 1.2. *Let $F \subseteq \mathbb{R}$ be a non-empty set. The following statements are equivalent:*

1. F is closed;
2. *if (x_n) is a sequence of points in F converging to some $x \in \mathbb{R}$, then $x \in F$.*

Proof. First suppose that F is closed, and let (x_n) be a sequence of points in F converging to some $x \in \mathbb{R}$. To establish 2., we must show that $x \in F$. Proceeding by contradiction, let us assume that $x \notin F$, i.e. $x \in F^c$. Since F is closed, F^c is open. In particular, it is an open set containing x . By the previous theorem, there is some $N \in \mathbb{N}$ such that $x_n \in F^c$ for all $n \geq N$. Obviously, this contradicts the assumption that (x_n) is a sequence of points in F . Therefore, we must have $x \in F$.

Conversely, suppose that criterion 2. holds true. By way of contradiction, suppose that F is not closed. This means that F^c is **not** open.³ Thus, there exists a point $x \in F^c$ such that, for each $\varepsilon > 0$, one has $V_\varepsilon(x) \not\subseteq F^c$. Equivalently, $V_\varepsilon(x) \cap F \neq \emptyset$ for each $\varepsilon > 0$. So, for $n \geq 1$, we can therefore choose a point

$$x_n \in F \cap V_{1/n}(x) = F \cap \left(x - \frac{1}{n}, x + \frac{1}{n}\right).$$

This gives us a sequence (x_n) living in F . We now claim that $\lim x_n = x$. Indeed, by the above:

$$|x_n - x| < \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

This means that (x_n) is a sequence of points in F converging to x . However, by the hypothesis in 2., this means that $x \in F$. Since $x \notin F$, we have reached our contradiction. Hence, F must be closed. $\square$

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³Although it may be tempting, F not being closed does **not** mean that F is open! Not closed does not imply open!